Operating System (25CS1302E)

Hospital Emergency Patient Monitoring System

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Under the Esteemed Guidance of

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Abstract

The Hospital Emergency Patient Monitoring System is a Linux-based project developed using C programming. It collects patient details such as Patient ID, heart rate, blood pressure, temperature, and oxygen level. The system uses Inter-Process Communication (IPC) to transfer patient information between two programs through FIFOs. The first program collects the patient details and sends them through patient_pipe. The second program receives the data and checks the patient's health condition. Based on the readings, the system classifies the patient as NORMAL, WARNING, or EMERGENCY. It displays the patient's condition and recommended action. For emergency cases, it displays a message about the need for immediate medical attention and doctor notification. The processed result is sent through hospital_fifo, and alert details are saved in alerts.log. This project helps us understand Linux processes, FIFO communication, system calls, and file handling through a simple hospital monitoring example.

Problem Statement & Objectives

Problem Statement

To develop a Linux-based system that collects patient vital signs, transfers the information between processes using FIFOs, identifies abnormal health conditions, and generates suitable alerts.

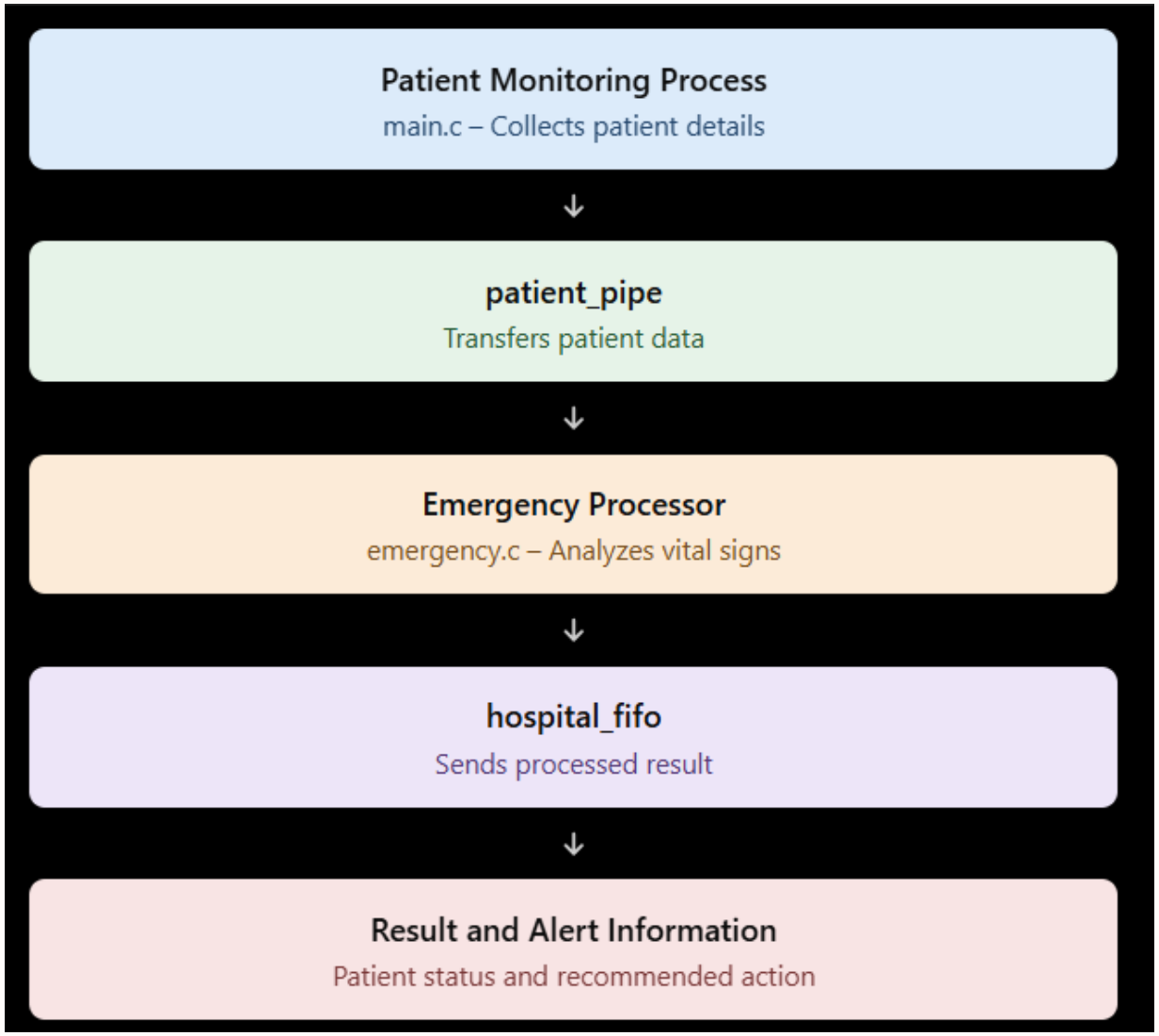
Objectives

* To simulate a hospital emergency patient monitoring system.
* To demonstrate Inter-Process Communication (IPC).
* To use pipes/FIFOs for communication between processes.
* To collect and process patient vital signs.
* To classify patient conditions into NORMAL, WARNING, and EMERGENCY.
* To display recommended actions and alert information.

Technologies Used

Technology	Purpose
C Programming	Develops the monitoring system
Linux / Ubuntu	Operating environment
GCC Compiler	Compiles C programs
VS Code	Writing and editing code
mkfifo()	Creates named pipes
open()	Opens FIFO communication channels
read() and write()	Transfers patient data
File Handling	Stores alerts in alerts.log

Flow Chart


Patient Monitoring Process
main.c – Collects patient details
patient_pipe
Transfers patient data
Emergency Processor
emergency.c – Analyzes vital signs
hospital_fifo
Sends processed result
Result and Alert Information
Patient status and recommended action

Patient Parameters and Status

Patient Parameters

Parameter	Description
Patient ID	Identifies the patient
Heart Rate	Measures heartbeats per minute
Blood Pressure	Records systolic and diastolic values
Temperature	Records body temperature
Oxygen Level	Records oxygen saturation

Patient Status

* NORMAL

Vital signs are within the normal range.

* WARNING

One or more vital signs are outside the normal range.

* EMERGENCY

Abnormal vital signs require immediate medical attention.

Implementation and Working Process

1. **Patient Data Collection:** main.c collects the Patient ID, heart rate, blood pressure, temperature, and oxygen level from the user.
2. **Data Transfer:** The patient details are stored in a PatientData structure and sent to the emergency processor through patient_pipe.
3. **Data Processing:** emergency.c receives the patient information and checks the vital signs against predefined health ranges.
4. **Status Classification:** Based on the readings, the system classifies the patient's condition as NORMAL, WARNING, or EMERGENCY.
5. **Result Display:** The system displays the patient's health status, detected condition, and recommended action.
6. **Alert Logging:** The processed patient details and alert information are stored in alerts.log for future reference.
7. **Result Communication:** The final processed information is sent through hospital_fifo for further communication.

IPC Concepts Used (Pipes and FIFOs)

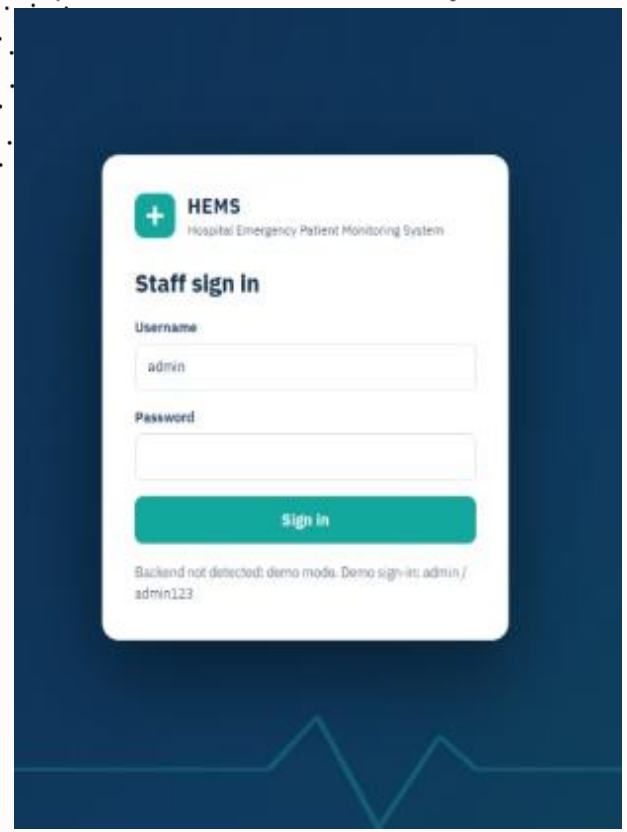
FIFO (Named Pipe):

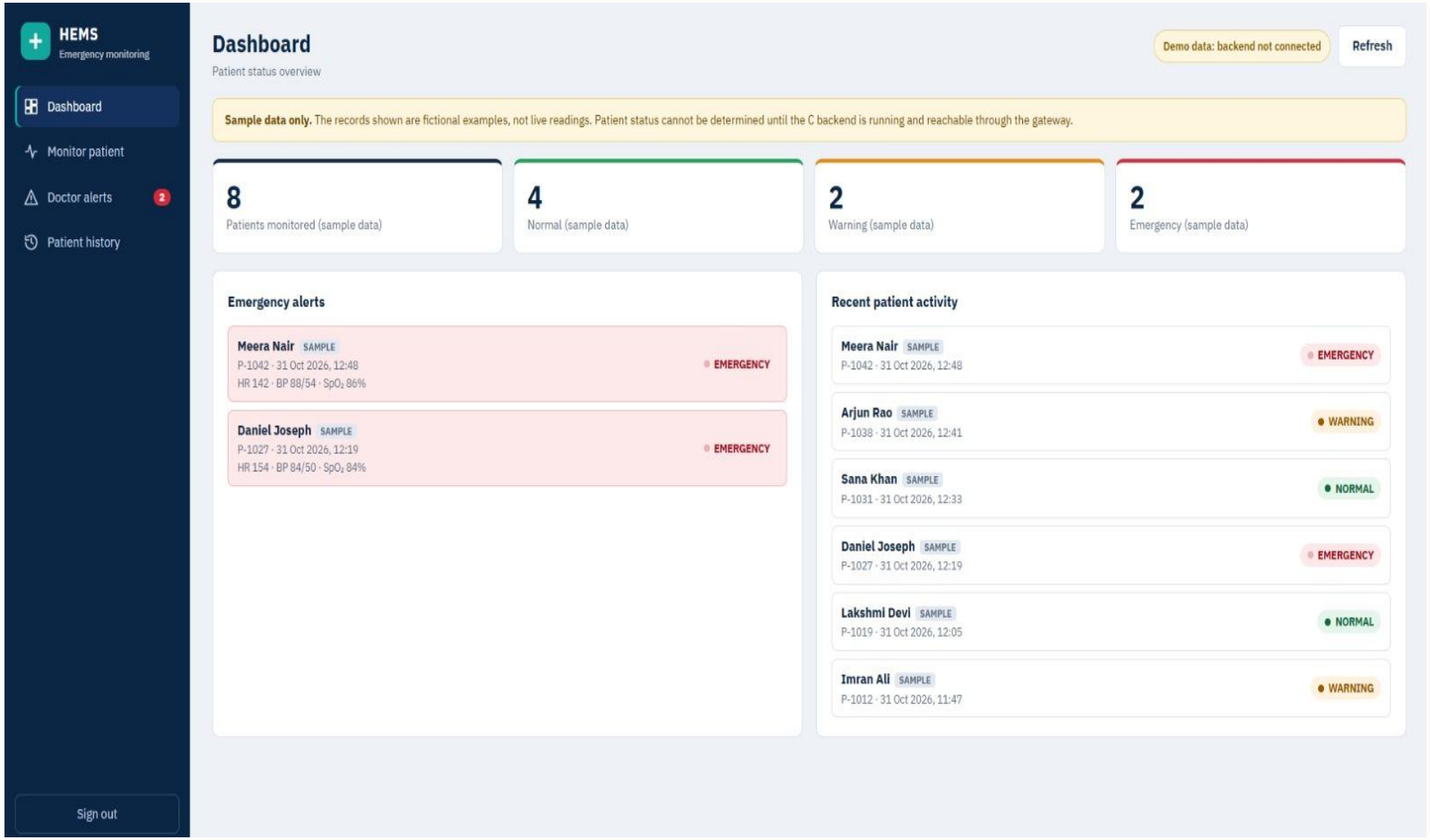
- Used to transfer data between the two C programs.
- patient_pipe transfers patient details.
- hospital_fifo transfers the processed result.
- FIFO communication allows one process to send information and another process to receive it.

Linux System Calls Used

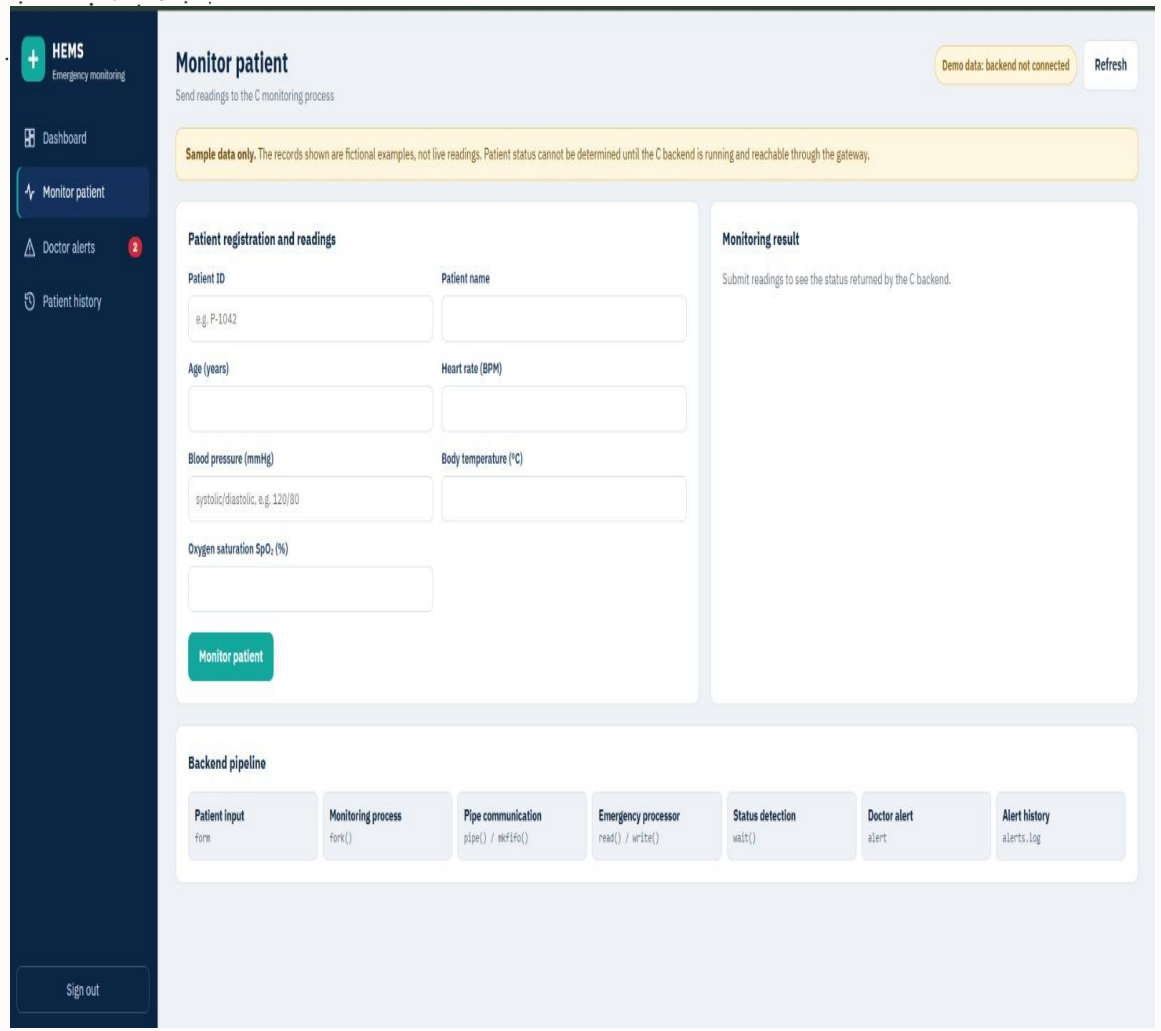
System Call	Purpose
mkfifo()	Creates a FIFO
open()	Opens a FIFO
read()	Reads patient data
write()	Sends patient data
close()	Closes the FIFO

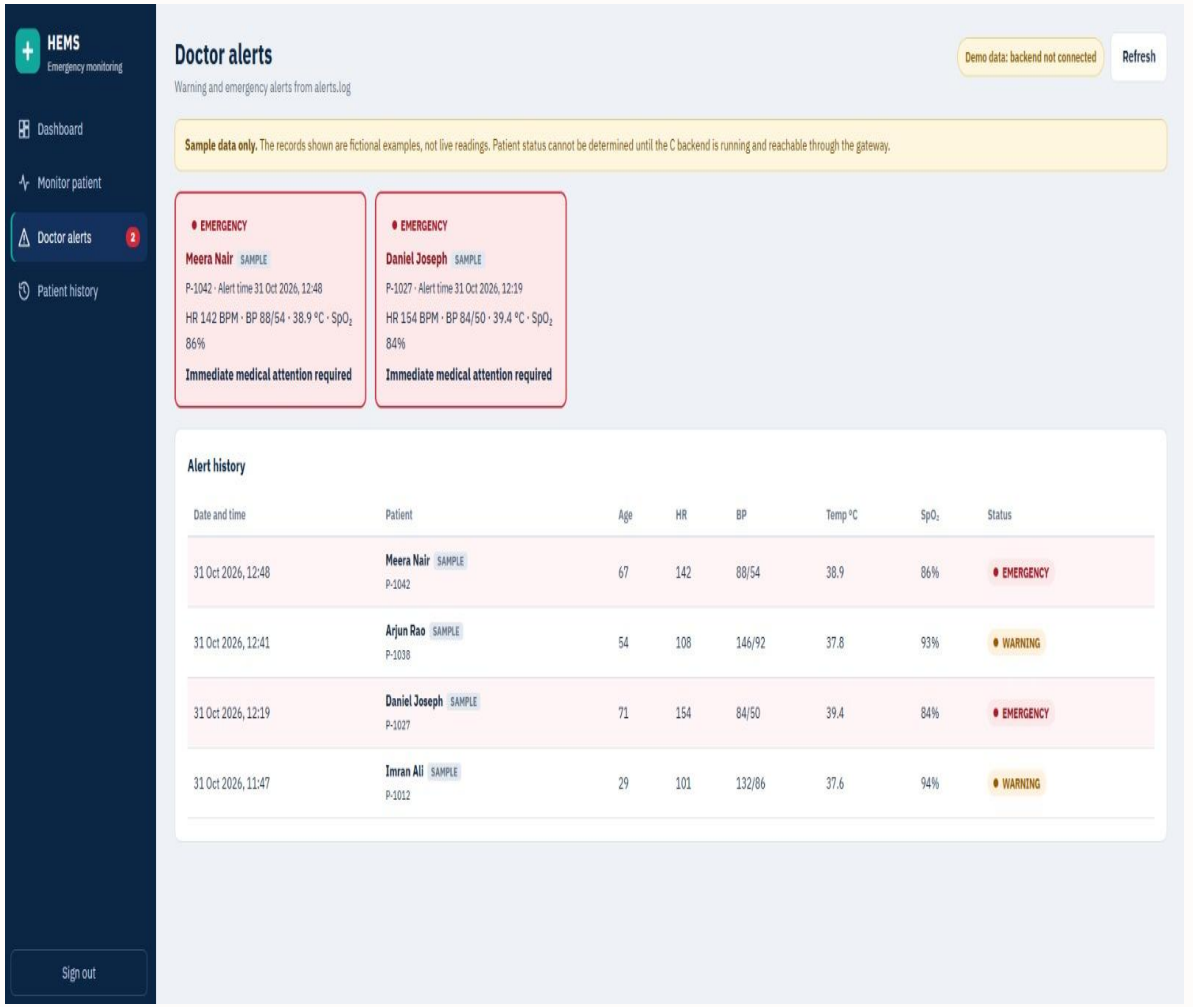
OutPut Screenshots





Screenshots





Expected output Future Scope

Expected Outcome

Successfully collects patient information.
Transfers data between Linux programs using FIFOs.
Checks patient vital signs.
Classifies patients as NORMAL, WARNING, or EMERGENCY.
Displays the condition and recommended action.
Stores alert details in alerts.log.

Future Scope

Connect the system to real patient monitoring devices.
Add a separate doctor notification process.
Store patient information in a database.
Develop real-time monitoring and notifications.
Create a web-based dashboard for doctors.

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Thank you